ScholarHub

A unified research workspace from *discovery to draft*.

SCHOLARHUB.SPACE

AUTHOR
Hassan Algizani
King Fahd University of Petroleum & Minerals

SUPERVISORS
Dr. Ahmad Irfan
Mr. Majed Alshaibani

PROGRAMME
MX Master's Programme
Final Project · 2026

APPROACH · SYSTEM

One project graph. Three workflows. Shared context throughout.

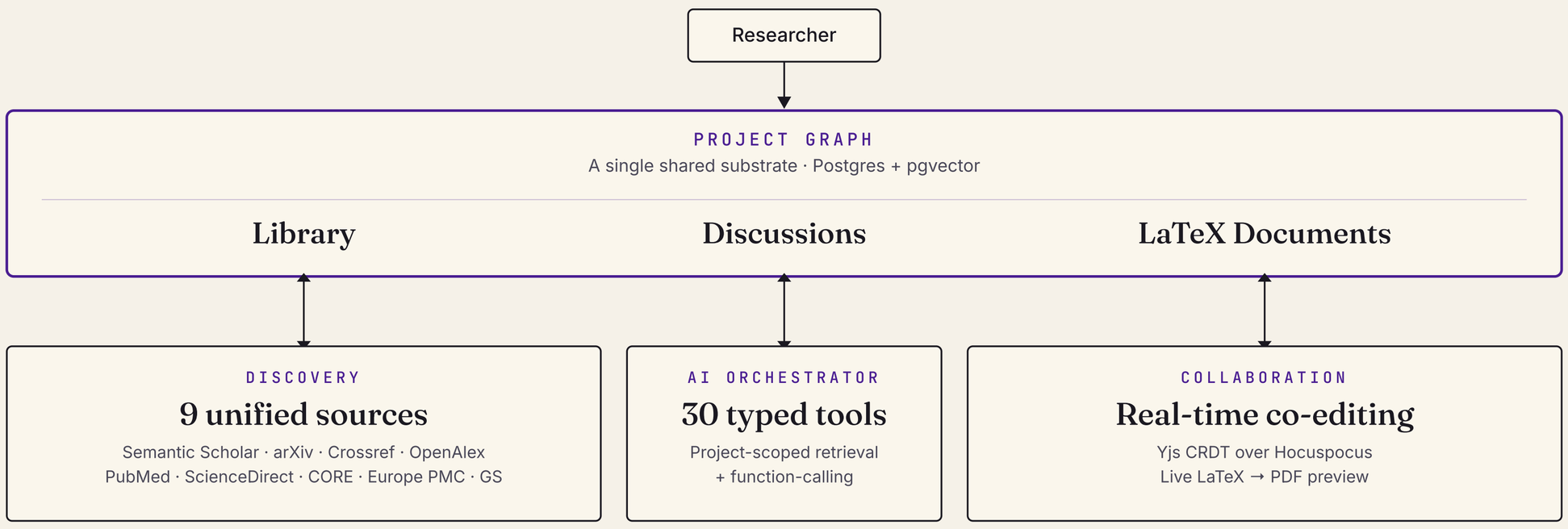
A single shared substrate holds the library, discussions, and LaTeX. Every component reads through the same protocol — so the AI's context is the project itself, not the open web.

01
Multi-Source Discovery
Nine academic databases, deduplicated into one project library.

02
Citation-Faithful Generation
AI cites only verifiable library identifiers — fabrication forbidden by protocol.

03
Real-Time Co-Editing
Multi-file LaTeX with live PDF preview, synchronised by Yjs CRDT.

SYSTEM DIAGRAM — THE PROJECT GRAPH AS SHARED RETRIEVAL SUBSTRATE



EVALUATION · SYSTEM-LEVEL VALIDATION

A working, publicly deployed system.

9
ACADEMIC SOURCES
Unified discovery layer

30
PROJECT-SCOPED TOOLS
Typed function-calling

0
INVENTED CITATIONS
By construction; protocol-enforced

CRDT
REAL-TIME COLLABORATION
Yjs over Hocuspocus

RESEARCH CONTRIBUTION

Three contributions of this thesis.

i. **Formalize** the project-graph as a shared retrieval substrate spanning library, discussions, and editor.

ii. **Constrain** AI generation to verifiable library identifiers, eliminating fabricated citations by construction.

iii. **Integrate** real-time collaborative LaTeX editing with project-scoped AI in one workspace.

FUTURE WORK

Open problems and next steps.

Controlled user study with graduate researchers, comparing citation accuracy and writing throughput against baseline tools.

Open the project-graph protocol to third-party reference managers and writing assistants.

Domain-specialised retrieval tuning (medical, ML, materials science).